

q1. what is react, and why is it popular?

React is popular JavaScript library is used for building user interfaces its key strength lies on ability to create component-based architecture which allow developers to create reusable UI components.

additionally react is virtual DOM which able to create copy of the real DOM that minimizes Direct DOM manipulation with significantly boosting performance

q2. What is a single page application(SPA)?

A single page application is a type of web application that loads a single html page and then instead of reloading the whole page only updates the affected part of the page is according to the user's action .

Q3. What is jsx, and how does it differ from HTML

Jsx or JavaScript XML is a syntax extension for JavaScript that allows developers to write html like code directly into JavaScript files.

Q4. Explain the difference between functional and class components?

Class components were used before hooks, class component were extended from the `react.component` and included the `render()` method which returned jsx .

State changes inside the components are managed using `this.state` and updated with `this.set state`.

Functional components being used more frequently after the introduction of react hooks use hooks like `useState` and `useEffect` to handle state and life cycle

Q5. What is difference b/w a stateless and stateful component in react?

Stateless component do not manage or store their data they simply receive data via prompts and display it. They are best used when you don't need to handle any logic and just want you content to displayed on the user interface.

Stateful components on other hand can manage their own state and can update their existing state based on user interaction or other events, this implies => that they can handle complicated logics and produce a required output as well

With the introduction of react hooks you can choose to keep functional components to be either stateful or stateless.

Q.5 what are props in react, and how are they used?

Props or properties are a way to pass data from parent component to a child component in react. The child components cannot modify these values because they are read only instead these values are used to trigger certain actions in the child components.

Q6. What is the difference between state and props in react?

State is a mutable object that stores dynamic data means it stores data that changes over time due to other factor such as user interaction network request or other event.

Props is an immutable objects means the data cannot be modified.

another notable difference is that state is private and is fully controlled by the component it belongs to

props is controlled only by the parent component and cannot be changed by the children

Q7. What are controlled and uncontrolled components ?

Controlled components are used when control is required over the data being entered into a form such as in the case of form validation where you need to fetch the username and password to validate their authenticity.

Uncontrolled components on the other hand are used when there is no need to dynamically inspect the user inputs.

Q.8 what is the purpose of key attribute in react lists?

Key attributes identifies each item uniquely when the list re-renders, enabling react to track and update individual items in the list without having to re-render the entire list

Q.9 what are fragments in react, and why are they useful?

Fragments in react is a way to group multiple elements together without adding extra nodes to the DOM, in short fragments are used in react when you want to avoid unnecessary wrapper element like div element.

Advance question in react interview

Q.10 what is the virtual DOM , and how does react use it to improve performance?

The virtual DOM is a lightweight copy or a lightweight in-memory representation of the the actual DOM.

Means=> when you are running a react component there are two doms one is original dom and one is virtual dom

1.why would we need a virtual dom

Whatever code you are writing it checks through or it scrolls through you whole code to see what has changed so it change only that individual elements which need the be changed with re render the whole page.

Q.11 what are react lifecycle methods, and when are they used?

React lifecycle methods are special functions that provide granular level control to developers to hook into specific points in a component's life cycle.

To attach certain event to a component's lifecycle is why you need the lifecycle methods

There are three main phases in the lifecycle

1.Mounting 2.updating 3.Unmounting

Mounted

- 1.constructor => initialize state, bind methods
- 2.getDerivedStateFromProp => update state based on props before rendering
- 3.render => return jsx for the component
- 4.componentDidMount => perform side effects (e.g. API calls)

Updating

- 1.getDerivedStateFromProp => update state based on props during updates
- 2.shouldComponentUpdate => decide if a re-render is necessary
- 3.render => return updated jsx for the component
- 4.getSnapshotBeforeUpdate => capture information before the DOM updates
- 5.componentDidUpdate => perform actions after the DOM update

Unmounting

- 1.componentWillUnmount => clean up resources, event listeners, subscriptions

Q.12 explain the useState and useEffect hooks with example?

useState is basically used to store the fetched data and the loading state

useEffect is used to fetch the data when the component mounts

Q.13 what do you understand by props drilling in react?

Props drilling is when data has to be passed down through many layers of components

Q.14 what is the Context API, and how is it used for state management?

Context API is a way to solve the problem of the props drilling how what context API does is instead of the props having to go through all of these intermediary components , context API can create a global variable or it can provide a global space where the these variables will

automatically get stored in so when any other child components wants to get this variable they don't have to ask their parent where is the variable and the parent ask their parent it does not have to be that way it can just directly go to this global space and retrieve the component or retrieve the variable whatever it is looking for.

In technical term the context api in react is a feature that allows you to manage and share state or data globally across a component tree, without needing to pass props manually at every level.

Syntax=>

```
Import React,{createContext,useState} from 'react'
```

```
Export const UserContext = createContext();
```

```
Export Const UserProvider= ({children})=>{
```

```
    Const [user,setUser] = useStae({name:'ehsan',age:22});
```

```
    Return(
```

```
        <UserContext.Provider value = {user}> {children}</UserContext.Provider>
```

```
    )}
```

Q.15 what are higher-order components?

It is a function that takes a components as an input and returns a new component with additional functionality or modified behaviour.

Q.16 explain the concept of reconciliation in react?

Reconciliation is like a process in react when it's like what happens in the background when a particular component is updated and how the virtual dom and the original dom is working together to display the content in user interface

There are five phases

- 1.initial render => when a component is first rendered, react creates a virtual DOM
- 2.state or props change=> when state or props change, react creates a new virtual DOM
- 3.diffing => react compares the old virtual DOM with new one
- 4.update the DOM => React updates only the parts of the actual DOM that need to be changed
- 5.commit phase => finally, react commits the changes to the actual DOM

Q.16 how do react portals work, and when should they be used?

In react portal is a way to render a component's children into a different part of the DOM outside the parent component's hierarchy, while maintaining the react component structure.

Q.17 how does react router handle navigation in single page app?

In single page application, react router is used to handle navigation between different views or "pages" without the need to reload the entire page

Q.18 what is react strict mode, and how does it help developers?

React strict helps developers to identify potential problems in an application by intentionally invoking certain warning and behaviour in the application.